RMO– 2018

Max Mark – 102 Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions.
- All questions carry equal marks.
- Answer to each questions starts on a new page. Clearly indicate the question number.

1. LET ABC be a triangle with integer sides in which AB<AC. Let the tangent to the circumcircle of triangle ABC at A intersect the line BC at D. Suppose AD is also an integer. Prove that gcd(AB, AC)>1.
2. Let n be a natural number. Find all real x satisfying the equation $\sum_{k=1}^{n} \frac{kx^n}{1+x^{2k}} = \frac{n(n+1)}{4}$.
3. For a rational number r, its period is the length of the smallest repeating block in its decimal expansion. For example, the number $r = 0.123123123...$ has period 3. If S denote the set of all rational number r of the form $r = 0.\overline{abcdefgh}$ having period 8, find the sum of all the elements of S.
4. Let E denote the set of 25 points (m, n) in the xy-plane, where m, n are natural numbers, $1 \le m \le 5,\ 1 \le n \le 5$. Suppose the point of E are arbitrarily coloured using two colours, red and blue. Show that there always exist four points in the set E of the form (a, b), (a+k, b), (a+k, b+k), (a, b+k) for some integer k such that three of these four points have the same colour. (That is, there always exists four points in the set E which form the vertices of a square and having at least three points of the same colour.)
5. Find all natural numbers n such that $1+[\sqrt{2n}]$ divides 2n. (For any real number x, $[x]$ denotes the largest integer not exceeding x.)
6. Let ABC be an acute angled triangle with AB<AC. Let I be the incentre of triangle ABC, and let D, E, F be the points st which its touches the sides BC, CA, AB respectively. Let BI, CI meets the line EF at Y, X respectively. Further assume that both X and Y are outside the triangle ABC. Prove that:
   (i) B, C, Y, X are concyclic;
   (ii) I is also the incentre of triangle DYX.

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